Chapter 5

The vector space R^n



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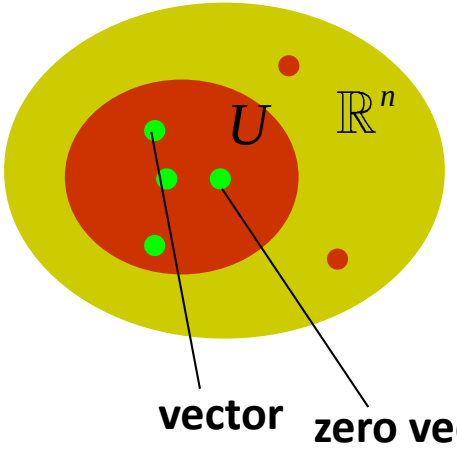
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5.1 Subspaces and Spanning sets

Subspace of R^n



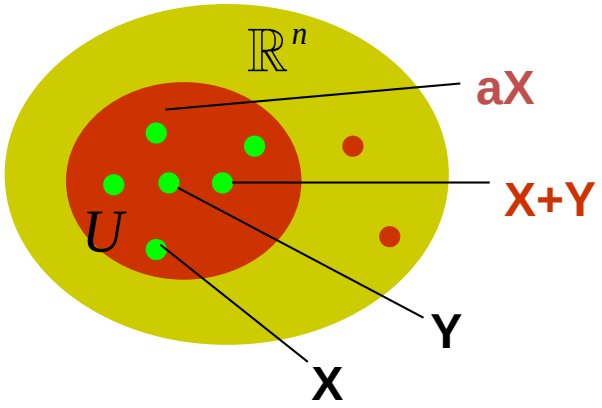
Definition of subspace of R^n.

- Let Ø≠U be a subset of R^n
- U is called a *subspace* of R^n if:

❶ S_1. The zero vector **0** is in U

❷ S_2. If **X**,**Y** are in U then **X**+**Y** is in U

❸ S_3. If **X** is in U then **aX** is in U for all real number **a**.



Subspace of R^n

- *Ex1*. $U=\{(a,a,0)|a\in R\}$ is a ***subspace*** of R^3
  - ❶ the zero vector of R^3, $(0,0,0)\in U$
  - ❷ $(a,a,0), (b,b,0)\in U \Rightarrow (a,a,0)+(b,b,0)=(a+b,a+b,0)\in U$
  - ❸ If $(a,a,0)\in U$ and $k\in R$, then $k(a,a,0)=(ka,ka,0)\in U$

- *Ex2*. $U=\{(a,b,1): a,b\in R\}$ is not a ***subspace*** of R^3
  - ❶ $(0,0,0)\notin U \Rightarrow$ U is not a ***subspace***

- *Ex3*. $U=\{(a,|a|,0)|a\in R\}$ is not a ***subspace*** of R^3
  - ❷ $(-1,|-1|,0), (1,|1|,0)\in U$ but $(0,2,0)\notin U \Rightarrow$ U is not a ***subspace***

- $V=\{[0\ a\ \ 0]^T \text{ in } \mathbb{R}^3: a \in Z\}$
- $U=\{[a\ \ 7\ \ 3a]^T \text{ in } \mathbb{R}^3: a \in R\}$
- $W=\{[5a\ \ \ b\ \ \ a-b]^T \text{ in } \mathbb{R}^3: a,b \in R\}$
- $Q=\{[a\ \ b\ \ |a+b|]^T: a \in \mathbb{R}\}$
- $H=\{[a\ \ \ b\ \ ab]^T: a,b \in \mathbb{R}\}$

Nhận xét: các trường hợp sau không là không gian vector con
- có thành phần khác không
- có hệ số bậc cao hoặc tích
- có dấu | |
- có a và a+1 chẳng hạn

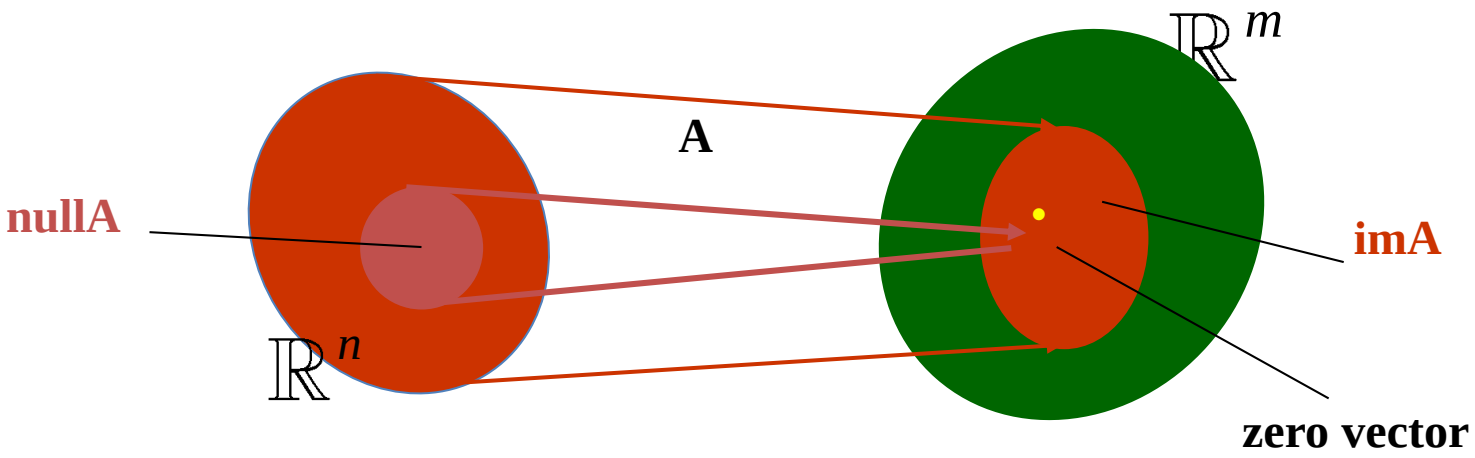
- $P=\{(x,y,z)|\ x-2y+z=0 \text{ and } 2x-y+3z=0\}$. P is called the *solution space* of the system $x-2y+z=0$ and $2x-y+3z=0$.

Note

- A subspace either has only one or infinite many vectors
- Example, {0} has only vector
- If a subspace U has nonzero vector X then aX is also in U (by S3). Then U has infinite many vector

Null space and image space of a matrix

- A is an mxn matrix, if X is nx1 matrix then AX is mx1 matrix
- **nullA** = {**X** in R^n: **AX=0**}
- **imA** = {**AX**: X is in R^n}



Null space and image space of a matrix

nullA $=\{X\in R^n: AX=0\}$ is a subspace of R^n:

❶ $A.0=0\Rightarrow 0\in nullA$

❷ $X,Y\in nullA \Rightarrow AX=0,\ AY=0$
$\Rightarrow A(X+Y)=AX+AY=0$

$\Rightarrow (X+Y)\in nullA$

❸ $X\in nullA,\ a\in R \Rightarrow AX=0$

$\Rightarrow A(aX)=a(AX)=0 \Rightarrow aX\in nullA$

imA $=\{AX: X\in R^n\}$ is a subspace of R^m:

❶ $0=A.0\Rightarrow 0\in imA$

❷ $AX,AY\in imA$
$\Rightarrow AX+AY=A(X+Y)=AZ$

$\Rightarrow AX+AY\in imA$

❸ $AX\in imA,\ a\in R$

$\Rightarrow a(AX)=A(aX)=AZ$

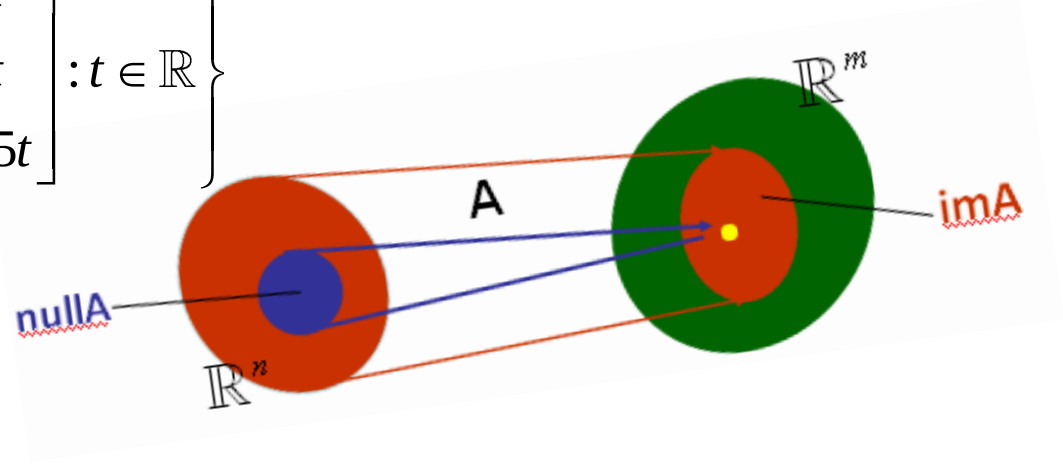
$\Rightarrow a(AX)\in imA$

Null space null *A={X:AX=0}*

- For example, $A=\begin{bmatrix}1 & -1 & 0\\ 2 & 3 & 1\end{bmatrix}_{2\times 3}$

$$nullA=\left\{\begin{bmatrix}x\\y\\z\end{bmatrix}:A\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}0\\0\end{bmatrix}\right\}=\left\{\begin{bmatrix}x\\y\\z\end{bmatrix}:\begin{bmatrix}1 & -1 & 0\\2 & 3 & 1\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}0\\0\end{bmatrix}\right\}$$

$$=\left\{\begin{bmatrix}x\\y\\z\end{bmatrix}:\begin{cases}x-y=0\\2x+3y+z=0\end{cases}\right\}=\left\{\begin{bmatrix}t\\t\\-5t\end{bmatrix}:t\in\mathbb{R}\right\}$$



Eigenspaces (không gian riêng)

- Suppose A is an nxn matrix and λ is an eigenvalue of A
- $E_\lambda(A)=\{X: AX=\lambda X\}$ is an subspace of $\mathbf{R}^n$
- For example,

$$A=\begin{bmatrix}-3 & -1\\ 0 & 2\end{bmatrix}\Rightarrow c_A(x)=\det(xI-A)=\begin{vmatrix}x+3 & 1\\ 0 & x-2\end{vmatrix}=(x+3)(x-2)$$

$$c_A(x)=0\Leftrightarrow x=-3\vee x=2$$

$$x=-3:\left[\begin{array}{cc|c}0 & 1 & 0\\ 0 & -5 & 0\end{array}\right]\rightarrow\left[\begin{array}{cc|c}0 & 1 & 0\\ 0 & 0 & 0\end{array}\right]\Rightarrow X=\begin{bmatrix}t\\ 0\end{bmatrix}\ (\text{or X}=(t,0))$$

$$x=2:\left[\begin{array}{cc|c}5 & 1 & 0\\ 0 & 0 & 0\end{array}\right]\Rightarrow X=\begin{bmatrix}t\\ -5t\end{bmatrix}$$

$$E_{-3}=\{X: AX=-3X\}=\{(t,0): t\in\mathbb{R}\}$$

$$E_2=\{X: AX=2X\}=\{(t,-5t): t\in\mathbb{R}\}$$

Các không gian riêng ứng với GTR

Spanning sets (hệ sinh)

- $Y=k_1X_1+k_2X_2+\ldots+k_nX_n$ is called a *linear combination* of the vectors $X_1,X_2,\ldots,X_n$
- The set of all *linear combinations* of the the vectors $X_1,X_2,\ldots,X_n$ is called the *span* of these vectors, denoted by $\text{span}\{X_1,X_2,\ldots,X_n\}$.
- This means, $\text{span}\{X_1,X_2,\ldots,X_n\} = \{k_1X_1+k_2X_2+\ldots+k_nX_n : k_i \in R \text{ is arbitrary}\}$
- $\text{span}\{X_1,X_2,\ldots,X_n\}$ is a subspace of R^n.
- For example, $\text{span}\{(1,0,1),(0,1,1)\}=\{a(1,0,1)+b(0,1,1) : a,b \in R\}$.
- And we have $(1,2,3) \in \text{span}\{(1,0,1),(0,1,1)\}$ because $(1,2,3)= 1(1,0,1)+ 2(0,1,1)$.
- $(2,3,2) \notin \text{span}\{(1,0,1),(0,1,1)\}$ because $(2,3,2) \neq a(1,0,1)+b(0,1,1)$ for all a,b.

Spanning sets (hệ sinh)

- Nếu U=span{X,Y} ta nói U là KG được sinh ra bởi {X,Y} hay hệ {X,Y} sinh ra KG U. Khi đó, U chứa tất cả các vector có dạng aX+bY với a, b là các số thực tùy ý.
- vector $Z \in span\{X,Y\}$ khi và chỉ khi có các số thực a,b sao cho Z=aX+bY hay hệ pt Z=aX+bY có nghiệm a,b.
- Ta cũng nói Z là một tổ hợp tuyến tính (linear combination) của X,Y khi Z=aX+bY hay $Z \in span\{X,Y\}$.

Examples

- If x=(1,3,-5) is expressed as a linear combination of the vectors $v_1 = (1, 1, 1)$; $v_2 = (1,1,-1)$; $v_3 = (1, 0, 2)$; then the coefficient of v_3 is:

A. 2 B. 3 C. -2 D. 1 E. 0

- x is expressed as a linear combination of v_1, v_2, v_3 means $x=av_1+bv_2+cv_3$ for some a,b,c and c is called the ***coefficient*** of v_3.
- the system is

$$\begin{aligned} a + b + c &= 1 \\ a + b &= 3 \\ a - b + 2c &= -5 \end{aligned}$$

1	1	1	1
1	1	0	3
1	-1	2	-5

1	1	1	1
0	0	-1	2
0	-2	1	-6

1	1	1	1
0	-2	1	-6
0	0	-1	2

$\Rightarrow a = 1$

$\Rightarrow b = 2$

$\Rightarrow c = -2$

Examples

- Which of the vectors below is a *linear combination* of u=(1,1,2); v=(2,3,5)?

A. (0,1,1) B. (1,1,0) C. (1,1,1) D. (1,0,1) E. (0,0,1)

- Có thể giải bằng biến đổi sơ cấp trên ma trận chứa các vector cột như sau:

u	v	A	B	C	D	E
❶	2	0	1	1	1	0
1	3	1	1	1	0	0
2	5	1	0	1	1	1

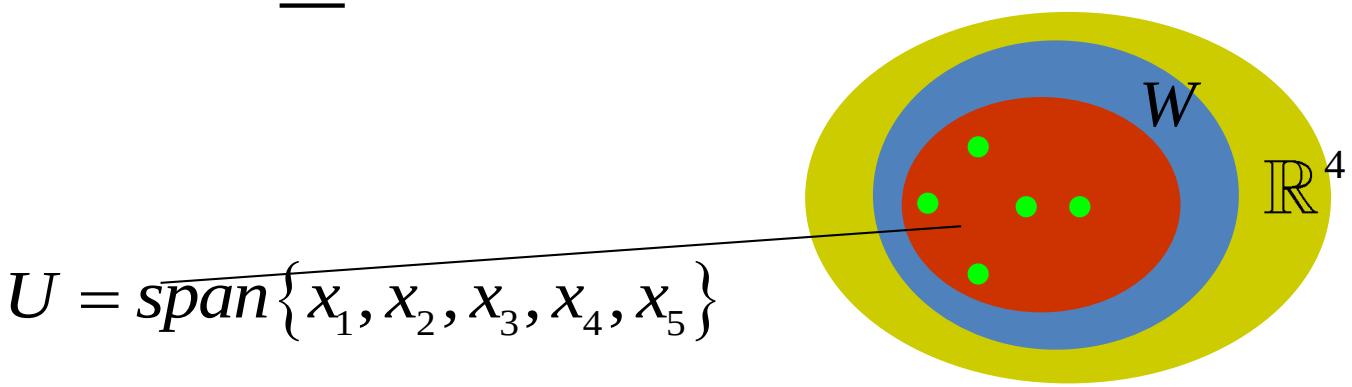
u	v	A	B	C	D	E
❶	-2	0	1	1	1	0
0	❶	1	0	0	-1	0
0	1	1	-2	-1	-1	1

u	v	A	B	C	D	E
❶	-2	0	1	1	1	0
0	❶	1	0	0	-1	0
0	0	0	-2	-1	0	1

Theorem

- *$U=span\{X_1,X_2,\ldots,X_n\}$ is in R^n and U is a subspace of R^n*
- *If W is a subspace of R^n such that X_i are in W then*

$$U \subseteq W$$



EXERCISES 5.1

1. In each case determine whether U is a subspace of $\mathbb{R}^3$. Support your answer.

   (a) $U = \{(1, s, t) \mid s \text{ and } t \text{ in } \mathbb{R}\}$.

   (b) $U = \{(0, s, t) \mid s \text{ and } t \text{ in } \mathbb{R}\}$.

   (c) $U = \{(r, s, t) \mid r, s, \text{ and } t \text{ in } \mathbb{R}, -r + 3s + 2t = 0\}$.

   (d) $U = \{(r, 3s, r - 2) \mid r \text{ and } s \text{ in } \mathbb{R}\}$.

   (e) $U = \{(r, 0, s) \mid r^2 + s^2 = 0, r \text{ and } s \text{ in } \mathbb{R}\}$.

   (f) $U = \{(2r, -s^2, t) \mid r, s, \text{ and } t \text{ in } \mathbb{R}\}$.

5.2 Independence and Dimension

Linear Independence (sự độc lập tuyến tính)

A set of vectors in R^m $\{X_1,X_2,\ldots,X_n\}$ is called **linearly independent** (độc lập tuyến tính)

if $t_1X_1+t_2X_2+\ldots+t_nX_n=0 \Rightarrow t_1=t_2=\ldots=t_n=0$ only

numbers in R vectors in R^m

Ex1. The set $\{[1\ \ -1]^T, [2\ \ 3]^T\}\subset R^2$ is called linearly independent since $t_1[1\ \ -1]^T + t_2[2\ \ 3]^T = [0\ \ 0]^T$ follows $t_1=t_2=0$.

Ex2. A set of vectors that containing zero vector never linearly independent.

Ex3. The set {(0,1,1), (1,-1,0), (1,0,1)} is *not linearly independent* because the system $t_1(0,1,1)+t_2(1,-1,0)+t_3(1,0,1)=(0,0,0)$ has one *solution $t_1=-1$, $t_2=-1$, $t_3=1$*

Examples

- Show that {(1,1,0);(0,1,1);(1,0,1)} is linearly independent in R^3

$$t_1(1,1,0)+t_2(0,1,1)+t_3(1,0,1)=(0,0,0)$$

$$\Rightarrow \ldots \Rightarrow t_1=t_2=t_3=0$$

$$t_1(1,1,0)+t_2(0,1,1)+t_3(1,0,1)=(0,0,0)$$

$$\Leftrightarrow \begin{cases} 1t_1+0t_2+1t_3=0 \\ 1t_1+1t_2+0t_3=0 \\ 0t_1+1t_2+1t_3=0 \end{cases} \Leftrightarrow \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 2 & 0 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right]$$

$$\Rightarrow t_1=t_2=t_3=0 \Rightarrow independent$$

Fast way to determine a set of vectors is independent or not:
independent ⇔ Number of leading 1s = member of vectors

More ex. {(1,0,-2), (2,1,0), (0,1,5), (-1,1,0)} is *not linearly independent* (number of leading 1s = member of vectors)

1	2	0	-1
0	1	1	1
-2	0	5	0

1	2	0	-1
0	1	1	1
0	4	5	2

1	2	0	-1
0	1	1	1
0	0	1	-2

Determine whether each the following sets is linearly independent or linearly dependent.

1. {(-1,2,0)}
2. {(0,0,0); (1,2,3); (-3,0,1)}
3. {(1,1,-1); (-1,1,1); (1,-1,1)}
4. {(-2,3,4,1); (4,-1,5,0); (-2,1,0,3)}
5. {(1,1,0); (-2,3,1); (5,0,1); (-1,0,2)}
6. {X-Y+Z,3X+Z,X+Y-Z}, where {X,Y,Z} is an independent set of vectors. (see below)

1	3	1
-1	0	1
1	1	-1

1	3	1
0	3	2
0	-2	-2

1	3	1
0	1	1
0	3	2

1	3	1
0	1	1
0	0	-1

1	3	1
0	1	1
0	0	1

⇒independent

Fundamental Theorem

- *Theorem.* Let U be a subspace of R^n is spanned by m vectors, if U contains k linearly independent vectors, then $k \leq m$
- This implies if $k>m$, then the set of k vectors is always linear dependence.
- For example, Let U be the space spanned by {(1,0,1), (0,-1,1)} and S={(1,0,1), (0,-1,1), (2,-1,3)} $\subset$U. Then, S is *not linearly independent* (*m=2, k=3*).

Basis and dimension (cơ sở và chiều của KG)

- ***Definition of basis***: Suppose U is a *subspace* of R^n, a *set* $\{X_1,X_2,\ldots,X_k\}$ is called a *basis* of U if $U=\text{span}\{X_1,X_2,\ldots,X_k\}$ and $\{X_1,X_2,\ldots,X_k\}$ is *linear independence*
- *Ex1*. Let $U=\{(a,-a)|a\in R\}$. Then U is a *subspace* of R^2. Consider the set $B=\{(1,-1)\}$. B is *linearly independent* and $U=\{(a,-a): a\in R\}=\{a(1,-1): a\in R\}=\text{span}\{(1,-1)\}$. So, B is a *basis* of U.
- Note that $B'=\{(-1,1)\}$ is also a *basis* of U.
- But $\{(1,1)\}$ is not a *basis* of U because U can not be spanned by $\{(1,1)\}$

Basis and dimension (cơ sở và chiều của KG)

- *Ex2*. Given that V=span{(1,1,1), (1,-1,0), (0,2,1)}. Then, B={(1,1,1), (1,-1,0), (0,2,1)} is not linearly independent, because (0,2,1)=(1,1,1) – (1,-1,0)⇒B is not a *basis* of V.
- Consider B'={(1,1,1), (1,-1,0)}. B' is *linearly independent* and all vectors in V are spanned by B' because

a(1,1,1)+ b(1,-1,0)+ c(0,2,1)

=a(1,1,1)+ b(1,-1,0)+ c(1,1,1) –c(1,-1,0)

= (a+c)(1,1,1)+(b-c)(1,-1,0). So, B' is a *basis* of V.

Some important theorems

- ***Theorem 1.*** The following are ***equivalence*** for an nxn matrix A.
  - ❶ A is invertible.
  - ❷ the columns of A are linearly independent.
  - ❸ the columns of A span R^n.
  - ❹ the rows of A are linearly independent.
  - ❺ the rows of A span the set of all 1xn rows.

Some important theorems

- ***Theorem 2.*** (Invariance theorem). If $\{a_1,a_2,..,a_m\}$ and $\{b_1,b_2,...,b_k\}$ are bases of a subspace U of R^n, then m=k. In this case, m=k is called ***dimension*** of U and we write ***dimU=m***.
- *Ex1*. Let $U=\{(a,-a)|a\in R\}$ be a subspace of R^2. Then, $B=\{(1,-1)\}$ is a *basis* of U and $B'=\{(-1,1)\}$ is also a *basis* of U. In this case, dimU=1.
- *Ex2*. $\{(1,0), (0,1)\}$ is a basis of R^2 and $\{(1,-2), (2,0)\}$ is also a basis of R^2. But $\{(1,0), (-1,1), (1,1)\}$ is not a basis of R^2. We have $dimR^2=2$.
- The basis $\{(1,0), (0,1)\}$ is called ***standard basis*** of R^2.
- *Ex3*. Which of the following is a basis of R^3?

  ❶ $\{(1,0,1), (0,0,1)\}$

  ❷ $\{(2,1,0), (-1,0,1), (1,0,1), (0,-1,1)\}$

  ❸ $\{(0,1), (1,0)\}$

  ❹ None of the others

Nhận xét: trong R^3, mọi tập có 3 vector độc lập đều là cơ sở.

- ***Theorem 3***. Let $U \neq 0$ be a subspace of R^n. Then:

❶ U has a basis and $\dim U \leq n$.

❷ Any independent set of U can be enlarged (by adding vectors) to a basis of U.

❸ If B spans U, then B can be cut down (by deleting vectors) to a basis of U.

Ex1. Let $U=\text{span}\{(1,1,1), (1,0,1), (1,-2,1)\}$ be a subspace of R^3. This means, $B= \{(1,1,1), (1,0,1), (1,-2,1)\}$ spans U.

❶ $\Rightarrow$U has a basis and $\dim U \leq 3$,

❸ $\Rightarrow$ B can be cut down to a basis of U: $\{(1,0,1), (1,1,1)\}$ is a basis of U, ***dimU=2***

❷ $\Rightarrow$construct a basis for U: $\varnothing \rightarrow \{(1,0,1)\} \rightarrow \{(1,0,1), (1,1,1)\}$.

1	1	1
1	0	-2
1	1	1

1	1	1
0	-1	-3
0	0	0

1	1	1
0	1	3
0	0	0

Some important theorems

- ***Theorem 4.*** Let U be a subspace of R^n and $B=\{X_1,X_2,\ldots,X_m\}\subset U$, where dimU=m. Then B is independent if and only if B spans U.

- ***Theorem 5***. Let $U\subseteq V$ be subspaces of R^n. Then:
  ❶ $\dim U \leq \dim V$.
  ❷ If dimU=dimV, then U=V.

3. Find a basis and calculate the dimension of the following subspaces of $\mathbb{R}^4$.

(a) span$\{(1, -1, 2, 0), (2, 3, 0, 3), (1, 9, -6, 6)\}$.

(b) span$\{(2, 1, 0, -1), (-1, 1, 1, 1), (2, 7, 4, 1)\}$.

(c) span$\{(-1, 2, 1, 0), (2, 0, 3, -1), (4, 4, 11, -3), (3, -2, 2, -1)\}$.

(d) span$\{(-2, 0, 3, 1), (1, 2, -1, 0), (-2, 8, 5, 3), (-1, 2, 2, 1)\}$.

Exercises

Determine whether U is a subspace of R^3.
$U=\{[0\ a\ b]^T: a,b \in R\}$
$U=\{[0\ 1\ s]^T: s \in R\}$
$U=\{[a\ b\ a+1]^T: a,b \in R\}$
$U=\{[a\ b\ a^2]^T: a, b \in R\}$

Nhận xét: không là subspace khi

- có hằng số khác 0
- có hai hệ số chênh lệch 1 hằng số
- có bậc lớn hơn 1

Exercises

Find all m such that the set {(2,m,1),(1,0,1),(0,1,1)} is linearly independent.
m≠-1
m=-1 only
m=0 only
m≠0
None of the others

1	0	2
0	1	m
1	1	1

❶	0	2
0	1	m
0	1	-1

❶	0	2
0	❶	m
0	0	-1-m

Independent ⇔ số ❶ = số vector

Exercises

A basis for the subspace $U=\{[a\ \ b\ \ a-b]^T: a,b \in R\}$ is…
a. $\{[1\ 0\ 1]^T, [0\ 1\ -1]^T\}$
b. $\{[1\ 1\ 0]^T\}$
c. $\{[1\ 0\ 1]^T, [-1\ 0\ -1]^T, [0\ 1\ -1]^T\}$
d. None of the others.

Nhận xét:

- U phụ thuộc 2 tham số nên dimU=2 và mọi cơ sở đều phải có đúng 2 vector độc lập tuyến tính$\Rightarrow$chỉ có thể là a hoặc d.
- kiểm tra a: độc lập và sinh ra U

Exercises

The dimension of the subspace U=span{(-2, 0, 3), (1, 2, -1), (-2, 8, 5),(-1, 2, 2)} is…
a. 2
b. 4
c. 3
d. 1

- không thể là b vì $\dim U \leq \dim R^3 = 3$
- kiểm tra bằng biến đổi sơ cấp

1	-2	-1	-2
2	8	2	0
-1	5	2	3

❶	-2	-1	-2
0	12	4	4
0	3	1	1

❶	-2	-1	-2
0	❶	1/3	1/3
0	0	0	0

có đúng hai ❶

Exercises

Let u and v be vectors in R^3 and $w \in span\{u,v\}$. Then …
a. {u,v,w} is linearly dependent.
b. {u,v,w} is linearly independent.
c. {u,v,w} is a basis of R^3
d. the subspace is spanned by {u,v,w} has the dimension 3.

- $w \in span\{u,v\}$ means $w=au+bv \Rightarrow$ {u,v,w} is not independent
- lưu ý cũng không có gì chắc chắn {u,v} độc lập nên $dimU \leq 2$

Exercises

Let {u,v,w,z} be independent. Then …. is also independent.
a. {u,v+w,z}
b. {u,v,v-z-u,z}
c. {u+v,u-w,z, v+z+w}
d. {u,v,w,u-v+w}

1	1	0	0
1	0	0	1
0	-1	0	1
0	0	1	1

❶	1	0	0
0	-1	0	1
0	-1	0	1
0	0	1	1

❶	1	0	0
0	❶	0	-1
0	0	❶	1
0	0	0	0

- {v-z-u,v,z,u} hiển nhiên phụ thuộc
- kiểm tra bằng biến đổi sơ cấp, ví dụ xét c, hệ số của các vector được đặt thành cột theo trật tự u,v,w,z
- chỉ có 3 leading ❶ trong khi có 4 vector⇒not independent

Exercises

- Let U=span{(1,-1,1), (0,2,1)}. Find all value(s) of m for which (3,-1,m)$\in$U.

- (3,-1,m) $\in$U$\Leftrightarrow$ (3,-1,m)= a(1,-1,1) + b(0,2,1) for some a,b. Solve for a,b$\Rightarrow$m=4

Exercises

- Find all values of m so that {(2,-1,3); (0,1,2); (-4,0; m)} spans R^3.

- ***Theorem 4.*** Let U be a subspace of R^n and $B=\{X_1,X_2,\ldots,X_m\}\subset U$, where dimU=m. Then B is independent if and only if B spans U.
- So, {(2,-1,3); (0,1,2); (-4,0; m)} spans R^3 $\Leftrightarrow$ it is linearly independent $\Leftrightarrow$ m≠10

2	0	-4
-1	1	0
3	2	m

1	-1	0
2	0	-4
3	2	m

1	-1	0
0	2	-4
0	5	m

1	-1	0
0	1	2
0	5	m

1	-1	0
0	1	2
0	0	m-10

Exercises

- Find a basis for the solution space to the homogeneous system

$$x - y + 2z = 0$$
$$2x + y + z = 0$$

1	-1	2	0
2	1	1	0

1	-1	2	0
0	3	-3	0

1	-1	2	0
0	**1**	-1	0

1	0	1	0
0	**1**	-1	0

Solution: $z=t$, $y=t$, $x=-t$

Solution space: $U=\{(-t,t,t) \mid t\in R\}$

$=\{t(-1,1,1) \mid t\in R\}=span\{(-1,1,1)\}$

A basis for U: $\{(-1,1,1)\}$

$dimU=1$

Exercises

- Find a basis for the subspace of R^3 defined by

$U=\{(a,b,c): 2a-b+3c=0\}$

a. {(1,2,0), (0,3,1)}
b. {(1,0,0), (0,1,0), (0,0,1)}
c. {(1,2,0)}
d. {(1,0,0), (1,2,0)}
e. {(3,0,-2), (1,0,0}
f. {(2,-1,3)}

Nhận xét:

- Nghiệm phụ thuộc 2 tham số$\Rightarrow$dimU=2 và mọi cơ sở của U phải có đúng 2 vector $\Rightarrow$chỉ a,d hoặc e đúng
- các vector trong cơ sở cũng phải thuộc U nên dễ thấy (1,0,0) không thuộc U$\Rightarrow$ loại d,e.

Exercises

- Find a basis and dimension for the subspace of R^3 defined by $U=\{(a,b,a-b) \mid a,b \in R\}$

- $U=\{a(1,0,1)+b(0,1,-1) \mid a,b \in R\}=span\{(1,0,1), (0,1,-1)\}$
- $\{(1,0,1), (0,1,-1)\}$ spans U and independent $\Rightarrow$ $\{(1,0,1), (0,1,-1)\}$ is a basis and dimU=2

Nhận xét:

- U phụ thuộc 2 tham số $\Rightarrow$ dimU=2
- Có thể chỉ ngay một cơ sở của U bằng cách bên (chọn theo cột)
- Còn nhiều cơ sở khác của U, ví dụ $\{(-1,0,-1), (0,1,-1)\}$,...

a	1	0
b	0	1
a-b	1	-1

- Find all values of m for which (1,2,m) lies in the subspace spanned by {(1,0,-1), (0,1,2)}
- Find all m such that (-3,2,m) is a linear combination of two vectors (1,1,-1) and (-2, 3, 4).
- Determine {x-y,x+y-z,x+z} is independent or not, where {x,y,z} is independent.

Determine whether each of the following sets is a basis of R^3 or not:

- {(1,0,-1), (0,1,2), (3,1,-1)}
- {(0,1,1), (-1,2,1)}
- {(1,0,0), (1,1,0), (1,1,1), (0,0,1)}

Find a basis and dimU if:

- U=span{(1,-1,2,0);(-2,1,0,1);(-1,0,0,1); (1,0,1,2)}
- U=span{(1,-1,3,0);(5,-2,4,3);(-2,0,7,1)}
- U=span{(-1,4,3);(3,0,-2);(-6,2,0)}
- U={(a,b,c):a+b+c=0}
- U={[a b 0]^T: a,b in R}

Subspace of R^n

- S1. The zero vector 0 is in U
- S2. x,y are in U → x+y is in U
- S3. x is in U → ax is in U for all real number a

→ U is a **subspace** of R^n

→ If (one of S1 or S2 or S3 is not true) then U is not a subspace of R^n

- U={(a,b,ab):a,b in R} is not a subspace
- U={(a,b,|a+b|): a,b in R} is not a subspace
- U={(a,b,0): a in Z, b in R} is not a subspace
- V={(a,b,2a-3b): a,b in R} is a subspace
- V={(a, b, c): a+b-c=0} is a subspace

Spanning set (hệ sinh)

- U=span{x,y}={ax+by:a,b in R}
- Vector z is in span{x,y} if and only if z is a **linear combination** of x and y, that means, there exist a and b such that z=ax+by
- U=span{x_1,x_2,…,x_n} is a subspace of R^n. Note that R^n=span{E_1,E_2,…,E_n}

Independence (sự độc lập)

- $\{x_1,x_2,\ldots,x_m\}$ is called linearly independent if

$$t_1x_1+t_2x_2+\ldots+t_mx_m=0$$

then

$$t_1=t_2=\ldots=t_m=0$$

- Note that if $\{x_1,x_2,\ldots,x_m\}$ is **linear independent** then every vector z in span$\{x_1,x_2,\ldots,x_m\}$ has a unique representation as a linear combination of x_i
- $\{x_1,x_2,\ldots,x_m\}$ is called **linearly dependent** (pttt) if it is not linear independent.

Basis and dimension

- If $\{x_1,x_2,\ldots,x_m\}$ is a basis of U then dimU=m
- $\dim R^n=n$
- Note that if U=span $\{x_1,x_2,\ldots,x_m\}$ then $\dim U\leq m$ and $\dim U=m$ if and only if $\{x_1,x_2,\ldots,x_m\}$ is linear independent
- If **dimU=m** then every set of **m+1** vector in U is linearly dependent
- Every set of n linearly independent vectors is a basis of R^n (mỗi tập gồm **n** vector **đltt** của $\mathbf{R^n}$ đều là một **cơ sở** của R^n)

Example

Which of the following are bases for $\mathbb{R}^3$?

(i) $\{(1,-2,0),(-2,1,3),(3,0,0)\}$

(ii) $\{(5,2,1),(0,1,-1),(3,0,2),(0,1,0)\}$

(iii) $\{(2,4,-1);(0,3,-1);(-4,-8,2)\}$

a	All of them
b	(i) only
c	(i) and (ii) only
d	(i) and (iii) only
e	None of them

Examples

- Find a basis and dimU if

U=span{(1,1,1);(1,-1,1);(-1,0,1);(0,-1,1)}

- Note that U is a subspace of R^3 , so $\dim U \leq \dim R^3 = 3$
- Which of these is a basis of R^3:

  A={(-1,0,3);(3,0,-1)};

  B={(1,-1,2);(3,0,1);(1,1,0)};

  C={(7,-1,4);(0,0,1);(1,-1,0);(0,1,5)};

Find all **x** in R such that {(1,1,1,1);(2,3,2,3),(3,4,1,**x**)} is a linearly independent set

Examples

- Find all ***x*** in R such that {(1,1,1,1);(2,3,2,3),(3,4,1,**x**)} is a linearly independent set
- Let U=span{(1,2,3);(3,4,5)}. Find **m** such that (3,5,***m***) lies in U
- For what value of **a** is the set of vectors S={(1,1,1,1); (3,2,1,5);(2,3,0,**a**-11)} is linearly dependent?A
- If (*x*,*y*,*z*) is expressed as a linear combination of vectors v1=(1,1,-1); v2=(1,0,1) and v3=(-1,0,1) then what is the coefficient of v3?

5.3 Orthogonality

Dot product

- If $X= [x_1 \ x_2 \ldots x_m]^T$, $Y= [y_1 \ y_2 \ldots y_m]^T$

We define

- $X \bullet Y = x_1y_1 + x_2y_2 + \ldots + x_my_m$

The length of a vector

- If $X=[x_1\ x_2\ \ldots\ x_m]^T$ then

$$\|X\| = \sqrt{X \bullet X} = \sqrt{x_1^2 + x_2^2 + \ldots + x_m^2}$$

- A vector with length 1 is called a unit vector
- **Distance** between X and Y defined by

$$d(X,Y) = \|X - Y\|$$

Theorem 1

Let X, Y and Z denote vectors in $\mathbb{R}^n$. Then:

1. $X \bullet Y = Y \bullet X$
2. $X \bullet (Y + Z) = X \bullet Y + X \bullet Z$
3. $(aX) \bullet Y = aX \bullet Y = X \bullet (aY)$ for all scalar a
4. $X \bullet X = \|X\|^2$
5. $\|X\| \geq 0$ for all vector X,
   $\|X\| = 0 \Leftrightarrow X = 0$
6. $\|aX\| = |a|\|X\|$ for all a

Example

$$(3\mathbf{x} - 4\mathbf{y}) \cdot (7\mathbf{x} + 2\mathbf{y})$$

$$= 21(\mathbf{x} \cdot \mathbf{x}) + 6(\mathbf{x} \cdot \mathbf{y}) - 28(\mathbf{y} \cdot \mathbf{x}) - 8(\mathbf{y} \cdot \mathbf{y})$$

$$= 21\|\mathbf{x}\|^2 - 22(\mathbf{x} \cdot \mathbf{y}) - 8\|\mathbf{y}\|^2$$

holds for all vectors **x** and **y**

Dot product, Length, and Distance

Example 3

Suppose that $R^n = \text{span}\{\mathbf{f}_1, \mathbf{f}_2, \ldots, \mathbf{f}_k\}$ for some vectors $\mathbf{f}_i$. If $\mathbf{x} \cdot \mathbf{f}_i = 0$ for each i where $\mathbf{x}$ is in R^n, show that $\mathbf{x} = \mathbf{0}$.

Solution ▶ We show $\mathbf{x} = \mathbf{0}$ by showing that $\|\mathbf{x}\| = 0$ and using (5) of Theorem 1. Since the $\mathbf{f}_i$ span $\mathbb{R}^n$, write $\mathbf{x} = t_1\mathbf{f}_1 + t_2\mathbf{f}_2 + \cdots + t_k\mathbf{f}_k$ where the t_i are in $\mathbb{R}$. Then

$$\begin{aligned}\|\mathbf{x}\|^2 &= \mathbf{x} \cdot \mathbf{x} = \mathbf{x} \cdot (t_1\mathbf{f}_1 + t_2\mathbf{f}_2 + \cdots + t_k\mathbf{f}_k)\\ &= t_1(\mathbf{x} \cdot \mathbf{f}_1) + t_2(\mathbf{x} \cdot \mathbf{f}_2) + \cdots + t_k(\mathbf{x} \cdot \mathbf{f}_k)\\ &= t_1(0) + t_2(0) + \cdots + t_k(0)\\ &= 0.\end{aligned}$$

Dot product, Length, and Distance

Theorem 2

Cauchy Inequality

If $\mathbf{x}$ *and* $\mathbf{y}$ *are vectors in* R^n, *then*

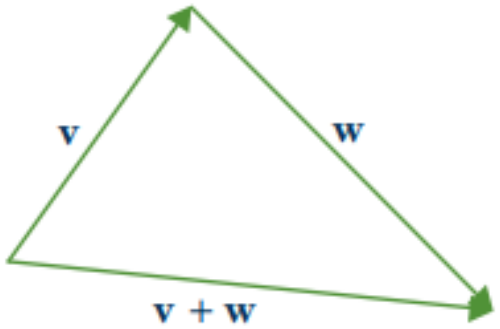
$$|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\|\|\mathbf{y}\|.$$

Moreover $|\mathbf{x} \cdot \mathbf{y}| = \|\mathbf{x}\|\|\mathbf{y}\|$ *if and only if one of* $\mathbf{x}$ *and* $\mathbf{y}$ *is a multiple of the other.*

Dot product, Length, and Distance

Corollary 1. Triangle Inequality

If $\mathbf{x}$ *and* $\mathbf{y}$ *are vectors in* $\boldsymbol{R}^n$, *then* $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$.



The reason for the name comes from the observation that in $\boldsymbol{R}^3$ the inequality asserts that the sum of the lengths of two sides of a triangle is not less than the length of the third side.

Dot product, Length, and Distance

Theorem 3

If $\mathbf{x}$, $\mathbf{y}$, *and* $\mathbf{z}$ *are three vectors in* $\boldsymbol{R}^n$ *we have:*

1. $d(\mathbf{x}, \mathbf{y}) \geq 0$ *for all* $\mathbf{x}$ *and* $\mathbf{y}$.
2. $d(\mathbf{x}, \mathbf{y}) = 0$ *if and only if* $\mathbf{x} = \mathbf{y}$.
3. $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$.
4. $d(\mathbf{x}, \mathbf{z}) \leq d(\mathbf{x}, \mathbf{y}) + d(\mathbf{y}, \mathbf{z})$. *Triangle inequality.*

Orthogonal Set (hệ trực giao)

- A set $\{x_1,x_2,\ldots,x_m\}$ is called **orthogonal set** if x_i is not zero vector and $x_i \bullet x_j = 0$ for all $i \neq j$
- For example, $\{(1,-1);(1,1)\}$ is an orthogonal set in R^2
- $\{(1,1,1);(-1,0,1);(0,1,0)\}$ is not orthogonal set but

$\{(-1,0,1);(0,1,0)\}$ is a orthogonal set.

- Note that $\{\mathbf{x}\}$ is an orthogonal set if $\mathbf{x} \neq \mathbf{0}$. A set $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k\}$ of vectors in $\mathbf{R}^n$ is called **orthonormal (hệ trực chuẩn)** if it is orthogonal and, in addition, each $\mathbf{x}_i$ is a unit vector: $\|\mathbf{x}_i\| = 1$ for each i.

Orthogonal Set (hệ trực giao)

For example,

- {(1,0,0);(0,1,0)} is orthonormal
- {(-3,0,4);(4,5,3)} is a orthogonal set, not a orthonormal set.
- However, the set

$$\left\{\frac{1}{5}(-3,0,4);\frac{1}{5\sqrt{2}}(4,5,3)\right\}$$

is orthonormal

Definition

Hence if $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k\}$ *is an orthogonal set, then* $\left\{\frac{1}{\|\mathbf{x}_1\|}\mathbf{x}_1, \frac{1}{\|\mathbf{x}_2\|}\mathbf{x}_2, \ldots, \frac{1}{\|\mathbf{x}_k\|}\mathbf{x}_k\right\}$ *is an orthonormal set, and we say that it is the result of* **normalizing** *the orthogonal set* $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k\}$.

Examples 6

If $\mathbf{f}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix}$, $\mathbf{f}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 2 \end{bmatrix}$, $\mathbf{f}_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$, and $\mathbf{f}_4 = \begin{bmatrix} -1 \\ 3 \\ -1 \\ 1 \end{bmatrix}$ then $\{\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3, \mathbf{f}_4\}$ is an orthogonal set in $\mathbb{R}^4$ as is easily verified. After normalizing, the corresponding orthonormal set is $\left\{\frac{1}{2}\mathbf{f}_1, \frac{1}{\sqrt{6}}\mathbf{f}_2, \frac{1}{\sqrt{2}}\mathbf{f}_3, \frac{1}{2\sqrt{3}}\mathbf{f}_4\right\}$.

Examples

- The standard basis of R^n $\{E_1,E_2,\ldots,E_n\}$ is orthonormal
- If $\{F_1,F_2,\ldots,F_k\}$ is orthogonal then $\{a_1F_1,a_2F_2,\ldots,a_kF_k\}$ is also orthogonal for any nonzero scalar a_i
- Every orthogonal set is a linearly independent set

Pythagoras's Theorem

- If $\{F_1,F_2,\ldots,F_k\}$ is orthogonal then

$$\|F_1+F_2+\ldots+F_k\|^2=\|F_1\|^2+\|F_2\|^2+\ldots+\|F_k\|^2$$

Expansion Theorem

- Let $\{F_1,F_2,\ldots,F_k\}$ be a orthogonal basis of a subspace U and X is in U. Then

$$X = \frac{X \bullet F_1}{\|F_1\|^2} F_1 + \frac{X \bullet F_2}{\|F_2\|^2} F_2 + \ldots + \frac{X \bullet F_n}{\|F_n\|^2} F_k$$

EXAMPLE 7

Expand $\mathbf{x} = (a, b, c, d)$ as a linear combination of the orthogonal basis $\{\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3, \mathbf{f}_4\}$ of $\mathbb{R}^4$ given in Example 6.

$$\mathbf{f}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix}, \mathbf{f}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 2 \end{bmatrix}, \mathbf{f}_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \text{ and } \mathbf{f}_4 = \begin{bmatrix} -1 \\ 3 \\ -1 \\ 1 \end{bmatrix}$$

5.4. Rank of a matrix

Rank of a matrix

- If A is carried to row-echelon form then rankA=number of leading 1's
- If A is an **mxn** matrix then **rankA≤min{n,m}**
- rankA=rank(A^T)

rowA and colA subspaces

- rowA=span{rows of matrix A}
- colA=span{columns of A}
- dim(rowA)=dim(colA)=rankA
- For example, find bases of colA and rowA if

$$A=\begin{bmatrix} 1 & 1 & -1 & 2 \\ 3 & 2 & 0 & 5 \\ -2 & -3 & 3 & -4 \\ 1 & 1 & -1 & 3 \\ 0 & 1 & -1 & 2 \end{bmatrix}$$

$$A=\begin{bmatrix}1&1&-1&2\\3&2&0&5\\-2&-3&3&-4\\1&1&-1&3\\0&1&-1&2\end{bmatrix}\rightarrow\begin{bmatrix}1&1&-1&2\\0&-1&3&-1\\0&-1&1&0\\0&0&0&1\\0&1&-1&2\end{bmatrix}$$

$$\rightarrow\begin{bmatrix}1&1&-1&2\\0&-1&3&-1\\0&0&-2&1\\0&0&0&1\\0&0&-3&3\end{bmatrix}\rightarrow\begin{bmatrix}1&1&-1&2\\0&-1&3&-1\\0&0&-2&1\\0&0&0&1\\0&0&0&0\end{bmatrix}$$

A basis of **rowA** is $\{r_1,r_2,r_3,r_4\}$ and **dim(rowA)=4**

A basis of **colA** is $\{c_1,c_2,c_3,c_4\}$ and **dim(colA)=4**

Theorem

- *An nxn matrix A is invertible if and only if rankA=n*
- *If an mxn matrix B has rank n then the n columns of B is linearly independent*
- *If A is mxn matrix and m>n then the set of m rows of A is not independent*

Theorem

If an mx**n** matrix A has **rank r** then

- The equation AX=0 has **n-r** basic solutions $X_1, X_2, \ldots, X_{n-r}$
- $\{X_1, X_2, \ldots, X_{n-r}\}$ is a **basis** of **nullA**
- Dim **nullA**=**n-r**
- **imA=colA** and
- Dim **imA**=dim **colA**=rankA=r

1. Obtain orthonormal bases of R^3 by normalizing the following.

(a) {(1, -1, 2), (0, 2, 1), (5, 1, -2)}

(b) {(1, 1, 1), (4, 1, -5), (2, -3, 1)}

4. In each case, write **x** as a linear combination of the orthogonal basis of the subspace *U*.

(a) **x** = (13, -20, 15);

U = span{(1, -2, 3), (-1, 1, 1)}

(b) **x** = (14, 1, -8, 5);

U = span{(2, -1, 0, 3), (2, 1, -2, -1)}

5. In each case, find all (a, b, c, d) in R^4 such that the given set is orthogonal.

(a) $\{(1, 2, 1, 0), (1, -1, 1, 3), (2, -1, 0, -1), (a, b, c, d)\}$

(b) $\{(1, 0, -1, 1), (2, 1, 1, -1), (1, -3, 1, 0), (a, b, c, d)\}$

6. If $\|\mathbf{x}\| = 3$, $\|\mathbf{y}\| = 1$, and $\mathbf{x} \cdot \mathbf{y} = -2$, compute:

(a) $\|3\mathbf{x} - 5\mathbf{y}\|$

(b) $\|2\mathbf{x} + 7\mathbf{y}\|$

(c) $(3\mathbf{x} - \mathbf{y}) \cdot (2\mathbf{y} - \mathbf{x})$

(d) $(\mathbf{x} - 2\mathbf{y}) \cdot (3\mathbf{x} + 5\mathbf{y})$

Examples

Find a basis and dimU if:

- U=span{(1,-1,2,0);(-2,1,0,1);(-1,0,0,1); (1,0,1,2)}
- U=span{(1,-1,3,0);(5,-2,4,3);(-2,0,7,1)}
- U=span{(-1,4,3);(3,0,-2);(-6,2,0)}
- U={(a,b,c):a+b+c=0}
- U={[a b 0]^T: a,b in R}

- P234: 1-4
- P244: 1-7
- P253: 1-6
- P261: 1-2